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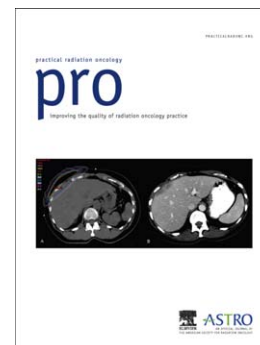
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Are we making an impact with incident learning systems? Analysis of quality improvement interventions using total body irradiation as a model system.

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ABSTRACT

Purpose

Despite increasing interest in incident learning systems (ILS) to improve safety and quality in radiation oncology, little is known about interventions developed in response to safety data. We used Total Body Irradiation (TBI) as a model system to study the effectiveness of interventions from our institutional ILS.

Methods and Materials

Near-miss event reports specific to TBI were identified from a departmental ILS from March 2012 to December 2015. The near-miss risk index (NMRI) was rated at multidisciplinary review from 0 (no potential harm) to 4 (critical potential harm). Interventions were analyzed for effectiveness with a schema adapted from The Joint Commission and other agencies: “Most Reliable” (e.g. forcing functions, automation), “Somewhat Reliable” (e.g. checklists, standardization), “Least Reliable” (e.g. training, rules, procedures). Causal factors of each event were drawn from the casual factor schema used in RO-ILS™.

Results

Of 4,007 safety-related reports, 266 reports pertained to TBI. TBI reports had a somewhat higher proportion of high-risk events (NMRI 3-4) compared to non-TBI reports (25% versus 17%, $p=0.0045$). A total of 117 interventions were implemented. The reliability indicators for the interventions were: Most Reliable (11% of interventions), Somewhat Reliable (17%), and Least Reliable (72%). Interventions were more likely to be applied to high-risk events (54% versus 41%, $p=0.03$). There was a pattern of high-reliability interventions with increased risk score of events. Events involving human error (e.g. slips) and equipment/IT lent themselves more often to

high-reliability interventions. Events related to communication, standardization, and training were associated with low-reliability interventions.

Conclusions

Over a three-and-a-half-year period, 117 quality improvement strategies were developed for TBI based on ILS. Interventions were more likely to be applied to high-risk events and high-risk events were more likely to be associated with high-quality interventions. These results may be useful to institutions seeking to develop interventions based on ILS data.

INTRODUCTION

Incident learning systems (ILS) define a process in which near-miss events or incidents are reported, analyzed for root causes, and appropriate interventions are performed to improve quality of care and patient safety. Interest in ILS to improve quality and safety in radiation oncology is increasing [1-3]. Studies have shown that ILS leads to identification of areas of weakness [4-6] and influence decision making processes for quality improvement (QI) interventions [7]. National incident reporting systems have been endorsed by various professional associations, including dedicated efforts and sponsorship of a national ILS by ASTRO and AAPM.

QI interventions aim to improve existing practices or prevent potential adverse events. Several studies in the healthcare field have evaluated outcomes of interventions. One example is an introduction of a physician computer order entry system into inpatient units leading to significant decline in preventable and non-intercepted potential adverse drug events [8]. There are also case studies of QI interventions in radiation oncology, such as modifying treatment software to prevent delivery of radiation in incorrect sequences [9] and a series of process modifications and introduction of new equipment to accommodate various patient setups in total body irradiation (TBI) [10]. Although these studies provide valuable information about QI interventions driven by ILS, these interventions are in response to relatively rare events that received a high level of attention, with mobilization of extensive labor and resources. Little is known about outcomes related to more routine use of a departmental ILS, i.e. day-to-day operations of an ILS which focus on near-miss events or process improvement and have a high volume of reporting. Key questions remain such as: how many events undergo QI interventions, what types of events

undergo QI interventions, what types of QI interventions are performed, and the outcomes of these interventions.

The goal of our study is to better understand QI interventions resulting from a departmental ILS by analyzing their associated causal factors, severity of events, and effectiveness. TBI was selected as a model system for several reasons. First, TBI is a high-volume procedure at our center (about 100 treatments per year) which provides a large database for study. Second, the techniques and technologies used within TBI are very homogeneous between patients. Third, the TBI workflow steps are broadly similar to those in general radiation therapy workflow (i.e. treatment planning, quality assurance, patient setups, and treatment delivery). Additionally, the TBI service contains many safety-critical aspects that are similar to those in general radiation therapy, such as communication between multiple professional staff within the department and communication between departments. As such, studying a complete model system offered a more complete analysis than focusing on single aspects of the radiation therapy workflow such as simulation or treatment planning.

METHODS

Our institutional ILS has been previously described [11] and this analysis was approved by the XXXX Institutional Review Board. The ILS was developed to evaluate departmental near-miss events, incidents, and process improvement proposals based on guidance from AAPM Working Group on the Prevention of Errors. Our radiation oncology staff are encouraged to report incidents, near-miss events, or quality improvement suggestions via a software tool on any departmental computer to facilitate a culture of safety. A multi-professional committee

consisting of physicians, physicists, dosimetrists, therapists, and nurses reviewed event reports on weekly basis. Our institution has a hospital-wide reporting system for the reporting of events that reach the patients that is used in combination with the ILS. Reported events were graded based on a near-miss risk index (NMRI), which has been previously described [12-13]. These are referred to as “risk scores” below, which ranges from 0 to 4 (none, mild, moderate, severe, critical, respectively), based on ease of detection and clinical consequences. During weekly ILS meetings, the multidisciplinary group discussed near miss events and came to consensus regarding causal factors, NMRI, and process improvement suggestions.

Reports specific to TBI were identified and from this subset, QI interventions in response to near-miss events were identified. The effectiveness of each intervention was categorized with a scheme adapted from The Joint Commission (Table 1): “Most Reliable” (e.g. forcing functions and automation), “Somewhat Reliable” (e.g. checklists, reminders, and standardization), and “Least Reliable” (e.g. education and training, rules, policies, and procedures) [14]. Causal factors, drawn from the RO-ILS causal factor schema, were identified for each event: communication within the radiation oncology department, communication outside the department (e.g. transplant service, affiliated hospitals), equipment or information technology (IT), human error (e.g. slip), standardization, training and education (defined as inadequate transmission of existing policy), and others (where the etiology of error could not be identified; e.g. patient status changes) [15]. Some events were associated with multiple causal factors. For statistical analysis, Excel was used for two sampled t-test with unequal variances to compare rates of near-miss events with high risk score (3-4) between TBI and non-TBI, as well as the rates of interventions between low risk (0-2) and high risk (3-4) TBI near-miss events. The level of significance was 0.05 for both analyses.

RESULTS

We analyzed 4,007 events from March 2012 to December 2015 from the departmental ILS. A search of the electronic medical record identified that 498 TBI patients were treated during this time period. Event reports in the ILS pertaining to TBI specific planning, treatment, and workflow processes were identified. In total 266 event reports related to TBI and 3,741 event reports not related to TBI (non-TBI) were identified.

Figure 1 shows the total number of events related to TBI versus all other events (non-TBI events) over time and the distribution of risk scores of TBI and non-TBI events during the same time period. The majority of both TBI events and non-TBI events were relatively low risk (scores 0-2). However, risk scores for TBI events were somewhat higher on average, including a greater portion of high risk events (scores 3-4) compared to that of the non-TBI events (25% versus 17%, $p=0.0045$).

In total, 117 QI interventions (44% of reported events) were implemented in response to near-miss events pertaining to TBI. Figure 2 shows the number of QI interventions based on risk score of events. Interventions were more likely to be applied to high risk score events – 41% of low risk (score 0-2) events led to interventions, compared to high risk (score 3-4) events, where 54% underwent interventions ($p = 0.03$).

We investigated the effectiveness category of the QI intervention (e.g. Least Reliable vs. Most Reliable). Table 1 shows example QI interventions and their corresponding effectiveness

category. An example of Most Reliable intervention is modifying the ruler so that only the metric units are visible and there can be no confusion with imperial units. An example of somewhat reliable intervention is creating a checklist for double checking measurements. An example of Least Reliable intervention is implementing a policy to make a direct phone call to a physician for late TBI schedule changes. The distribution of effectiveness category of QI interventions was as follows: 11% of interventions were Most Reliable, 17% Somewhat Reliable, and 72% Least Reliable. There was a pattern that events with higher risk scores were associated with higher reliability interventions, although this was not statistically significant (Figure 3).

In our analysis of causal factors that drive error, the most common causal factors of TBI events were: communication within radiation oncology department (39.4%), human error (27%), standardization (18.8%), communication outside the department (16.5%), equipment/IT (10%), and training and education (4%). Some of these causal factors were associated with more interventions than others: equipment/IT (59% of events in this category had QI interventions), standardization (72%), and training (91%) (Table 2). An example of equipment/IT related intervention was transitioning from paper to electronic records for TBI measurements with added function for automated error checking (see Table 1).

We also investigated the effectiveness category of the QI intervention as a function of the causal factor category (Table 2). Some events were associated with multiple causal factors: 39 incidents were associated with 2 causal factors and 2 incidents were associated with 3 causal factors. Most Reliable interventions tend to occur for events involving human error (19%) and equipment/IT (31%). Least Reliable interventions were associated with training (90%),

intradepartmental communication (83%), standardization (81%), and interdepartmental communication (69%).

DISCUSSION

While there is increasing interest in incident learning systems for safety and quality improvement in radiation oncology, little is known about how QI interventions are developed from incident learning data. We analyzed the QI interventions from a departmental ILS by analyzing their associated causal factors, severity of events, and effectiveness. TBI was selected as a model system due to the large amount of data available and the generalizability of the overall workflow and safety critical aspects to more general radiation oncology service.

Even for a fairly standardized procedure like TBI, we found that nearly half of the treated TBI patients had near-miss events reported in the ILS. The majority of TBI events were of low risk (score 0-2), although the high risk events (score 3-4) showed a modest but significant increase over non-TBI events. The high proportion of low risk events is similar to the distribution of reporting for the entire database, as individuals are encouraged to report even minor events which represent opportunities for process improvement. Additionally, the number of TBI events over time showed a similar trend to non-TBI events over time, suggesting that TBI was a reasonable model system.

Our analysis identified 117 QI interventions applied to the TBI process and procedures in response to event reports. The number of identified interventions is encouraging, as it demonstrates that QI interventions can be developed from ILS, which may be useful as an

administrative, accreditation, or academic metric of success. It also reveals challenges of developing appropriate interventions to every event in a high-volume ILS which receives over 1000 reports per year. QI interventions were more likely to be applied to events with high risk scores. It is intuitive that higher-risk events were associated with greater QI efforts (for example, high risk events may undergo formal root cause analysis which may reveal potential interventions) as the potential consequences are more severe by definition. Many high risk near-miss events that did not lead to clear interventions originated from human errors and communication issues (e.g. missing an email notification or miscommunication with patient transport). Because many communication issues involved communication outside of the department, it was particularly challenging to implement systematic changes.

Some QI interventions are considered more reliable than others (e.g. a mechanical interlock vs. an email reminder; see Table 1). In this study, the majority of QI interventions were in the Least Reliable category (72%), which consists of policy and procedural changes, and education and training of involved staff. We also found that more reliable QI interventions were associated with higher-risk events. This is understandable, as more reliable interventions tend to introduce systems that require specific skills and resources (e.g. fixing bugs in treatment planning software, creating alerts when out of range values are entered into the system, adding a new function for automatic data transfer), as opposed to low-reliability interventions (e.g. creating a check list or sending out reminder emails). Therefore, reliable but labor-intensive QI interventions tend to be reserved for the most critical near-miss events.

Given the reliance on the Least Reliable category of intervention, it is of interest to investigate

ways to improve these QI interventions. For example, one QI intervention in this study focused on communication about compensator changes to TBI treatments. A policy was put into place as to how such changes should be documented and communicated. However, there are different ways to implement such a policy change, each of which has a variable level of effectiveness. For example, an informal announcement to individual staff is quite different than an all-staff meeting with formal presentations of the policy changes.

In this study we also looked at causal factors associated with QI interventions. IT/equipment-related events were associated with high rates on QI interventions, and these interventions tended to fall in the Most Reliable category. This perhaps reflects how effectiveness is defined, which in general involves elimination of manual processes [7, 16 -17]. We also found that casual errors in standardization led to high rates of somewhat reliable intervention (which includes standardization), and causal errors in training led to high rates of least reliable interventions (which includes education and training). Perhaps the match between the causal factors and the interventions employed is related to the ease of identification and implementation of appropriate interventions. Given the relatively large number of variables (6 causal factors with 3 intervention levels), multivariate analysis was not performed in the current study.

Acknowledging the potential associations between intervention effectiveness and causal factors, however, multivariate studies of causal factors and intervention effectiveness are an interesting direction for future multicenter studies.

Notably, inadequate communication was the most frequent factor leading to errors and was also most frequently associated with low rates of interventions and lower quality interventions. This

reflects particular challenges in addressing communication in the modern era such as difficulty establishing a streamlined communication tool for various staff within and between departments, variable availability of staff, different levels of expertise and expectations, and evolving treatment techniques. In fact, these problems are not just unique to TBI patients or radiation oncology patients but also to healthcare in general [18-21]. This underscores the importance of developing communications tools to coordinate efficient care.

One may note that there is no significant change in the number or severity of events seen over time, and question whether ILS leads to improvement of the work flow and safety. However, this cannot be used to quantify process improvement for a number of reasons. First, the total number of reported events over time increased with increased awareness and staff participation in ILS [7, 13]. Thus a continuously decreasing event report rate is not necessarily a desirable outcome. Second, an intervention itself can generate multiple downstream issues [7]. For example, there were multiple incident reports for missing appointments for on treatment visits (OTV) for TBI despite multiple interventions. Some of the challenges included identifying patients for an OTV in timely fashion when treatment schedules are abbreviated, finding physicians for weekend treatment coverages, and adjusting OTV schedules when late changes occur to the treatment plans. Multiple interventions were done over the years, such as introducing an algorithm to automatically capture missing appointments in the scheduling software and clarifying ownership of TBI scheduling by reassigning scheduling responsibility from three staff groups to a single staff group. After multiple trial-and-errors, reflected by continuing feedback through incident reporting, the scheduling has become less error-prone. Again, this example demonstrates that the number of ILS reports for a given near-miss event (e.g. missing OTV appointment) would not

necessarily decrease immediately after an intervention, although the causal factors (e.g. communication error, policy changes, training, IT) or risk score may change during the process of improvement.

One of the limitations of the study was that the data collection and analysis of the intervention were done retrospectively, and description of interventions frequently lacked detail. It is also possible that there were unrecorded interventions. In our institutional ILS, a free text box is used to describe, analyze, and respond to events. Documentation of interventions in institutional ILS or national ILS such as RO-ILS may be improved by adding structured data fields specific to interventions. For example, reports without intervention data entered within two weeks could be flagged for further review in the system. It may also be of interest to reserve time in safety meetings dedicated to reviewing the status of proposed interventions. Additionally, failure mode and effects analysis (FMEA) has been shown to be a useful complement to ILS and the application of FMEA to ILS interventions is an interesting future direction [25].

Another limitation of the study is the three category classification of intervention reliability. While we adapted this from other work [15], more effort is needed to refine and validate intervention effectiveness. In fact, although vast majority of interventions were in the Least Reliable category, it is possible that these interventions were adequate for given near-miss events. Some studies have suggested a way to quantify this through a “solution priority number,” which accounts not only for effectiveness of an intervention, but also for the cost and ease of its implementation [16, 22]. Quantification of these factors would require more long-term data, expertise, and resource, but would be an interesting area of research. Additionally, individuals

are subject to rater bias in assigning intervention effectiveness and the reliability of scoring between different raters is unknown. There is a limited body of work on interrater reliability of risk scores for near miss incident reporting [23] and FMEA [24] which suggest that near miss events can be classified with good agreement relative to other formalisms such as the Agency for Healthcare Research and Quality (AHRQ) Harm Score. Evaluating reliability of interventions will be an important future direction as consensus views of classification for incident learning are developed.

CONCLUSION

The current analysis has examined the link between ILS and QI interventions. Interventions were more likely to be applied to high-risk events and high-risk events were more likely to be associated with high-quality interventions. These results may be useful to institutions seeking to develop interventions based on ILS data. These results underscore the need for formal recording, tracking, and evaluation of the interventions as a part of ILS.

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FIGURE AND TABLE LEGENDS

Figure 1. Number of events reported to the ILS classified as TBI and non-TBI over the study period (top row) and TBI and non-TBI event risk scores distribution (bottom row). The distribution of reports was similar between TBI and non TBI events, with some increase in average risk score associated with TBI events.

Figure 2. QI intervention counts per risk score of events. Higher risk events were associated with greater rates of interventions.

Figure 3. Risk scores of events and reliability of QI interventions.

Table 1. Intervention examples in TBI near-miss events based on The Joint Commission reliability schema.

Table 2: QI intervention counts and effectiveness category of interventions per causal factors.

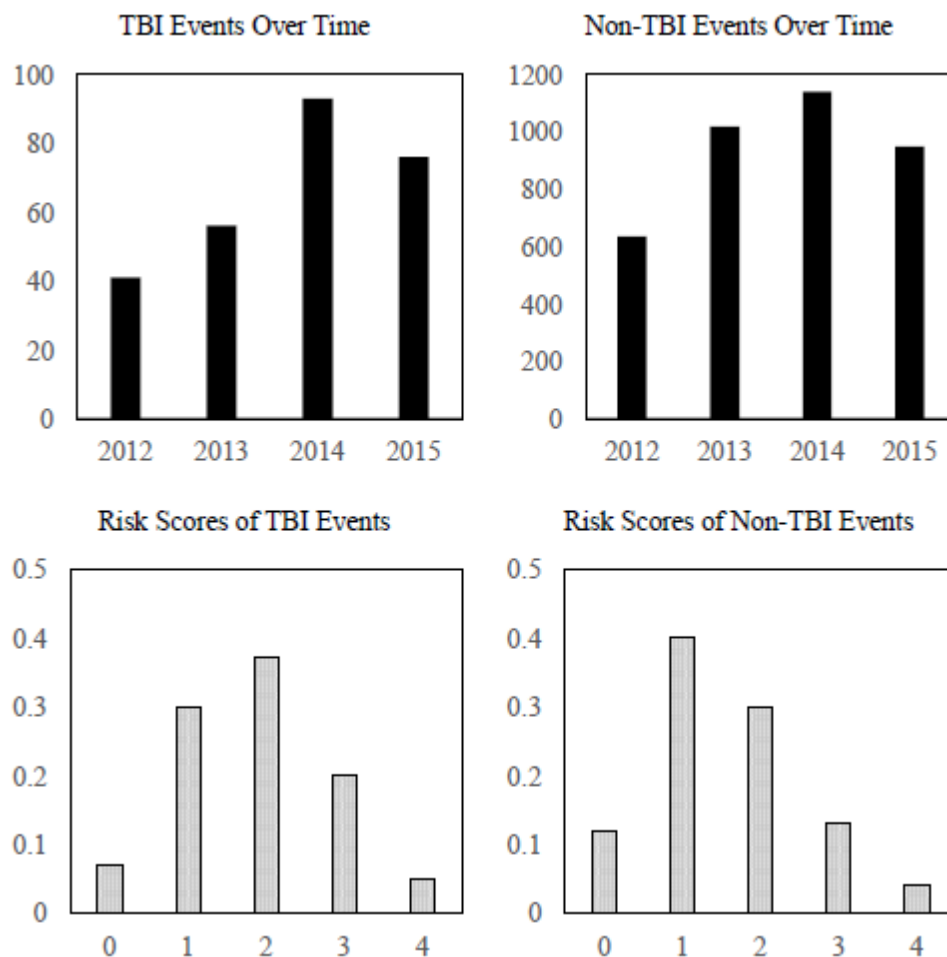


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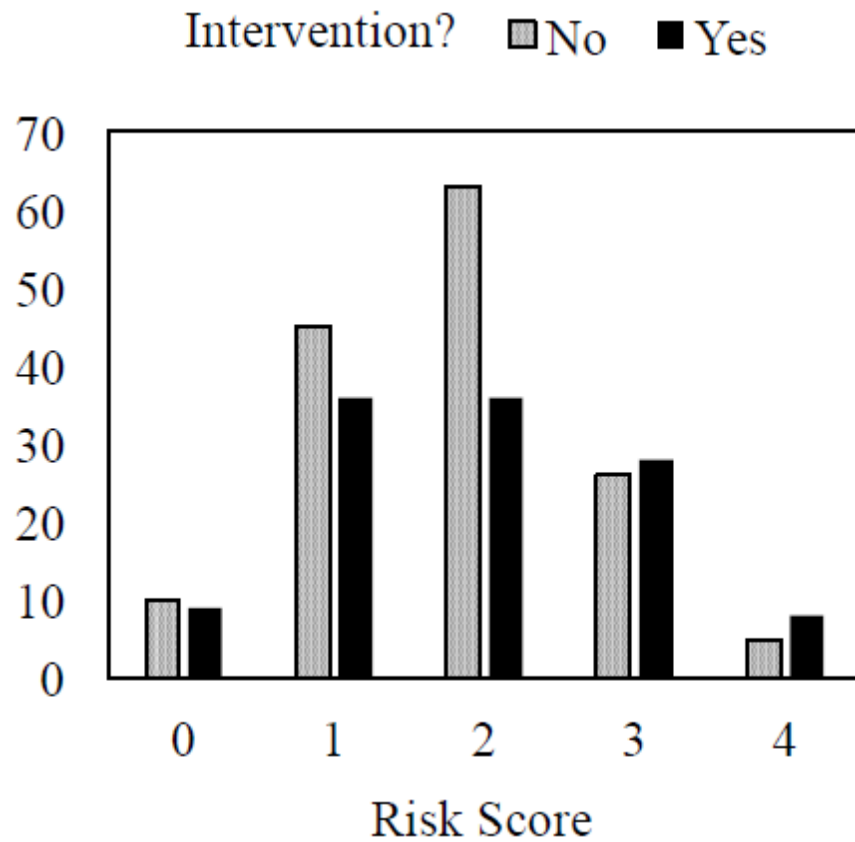


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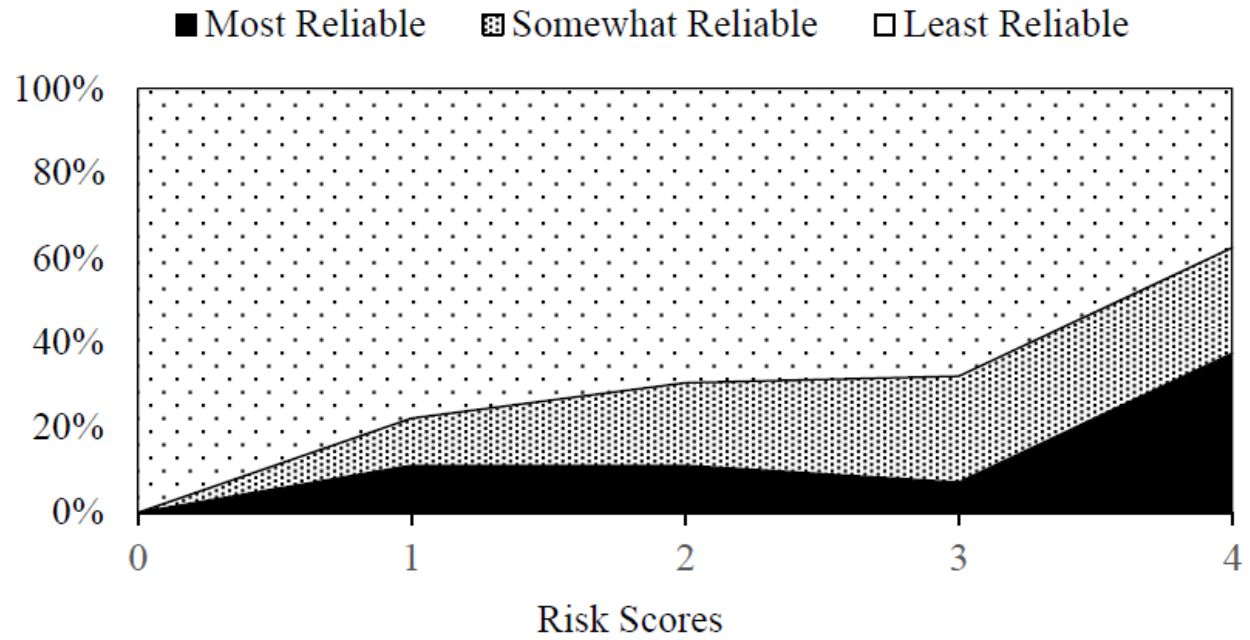


Figure 3. Risk scores of events and reliability of QI interventions.

Table 1. Intervention examples in TBI near-miss events based on Joint Commission reliability schema

	Hierarchy of Effectiveness Strategies	Example event	Example Intervention
Most Reliable	- Forcing functions or physical stops that prevent incorrect actions - Computerized automation - Human-machine redundancy	- Wrong TBI dose calculation due to having incorrect measurements from using a ruler with both imperial and metric units - Wrong TBI dose calculations due to a wrong dose entered into a paper form	- Placing a tape around the ruler, making only the metric units visible - TBI measurements entered into electronic records with added function for automated error checking.
Somewhat Reliable	- Reminders - Checklist - Forced pause to recheck details and steps - Planned error-recovery opportunities	- Missing diode measurement due to unclear staff assignment over the weekend - Wrong TBI dose calculations from inaccurate knee and ankle measurements	- Staff assignments to be emailed to on-call staff prior to weekend - Checklist added for double checking measurements
Least Reliable	(Enforcement of) Rules, policies, and procedures	- Late cancellation of the TBI appointment was communicated via email to a physician, which was not found until the day of cancelled appointment - Patient unable to	- Policy to make a direct phone call to a physician for late schedule changes - New policy to have

	• Education and training	stand for TBI treatment required urgent decubitus replanning, leading to delays • Incorrect TBI block tray code when the TBI system was transitioned from using acrylic compensators (on the beam spoiler) to lead compensators (on the linac tray).	both standing and decubitus calculations performed prior to initial treatment. • Educated the physicist and therapists about the new compensator
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Table 2: QI intervention counts and effectiveness category of interventions per causal factors

	Total Events	Total Interventions	%	Most Reliable Interventions	%	Somewhat Reliable Interventions	%	Least Reliable Interventions	%
Intradepartmental communication	105	42	40	2/42	5	5/42	12	35/42	83
Interdepartmental communication	44	16	36	1/16	6	4/16	25	11/16	69
Equipment/IT	27	16	59	5/16	31	1/16	6	10/16	63
Human error	72	26	36	5/26	19	11/26	42	10/26	38
Standardization	50	36	72	1/36	3	6/36	17	29/36	81
Training	11	10	91	0/10	0	1/10	10	9/10	90